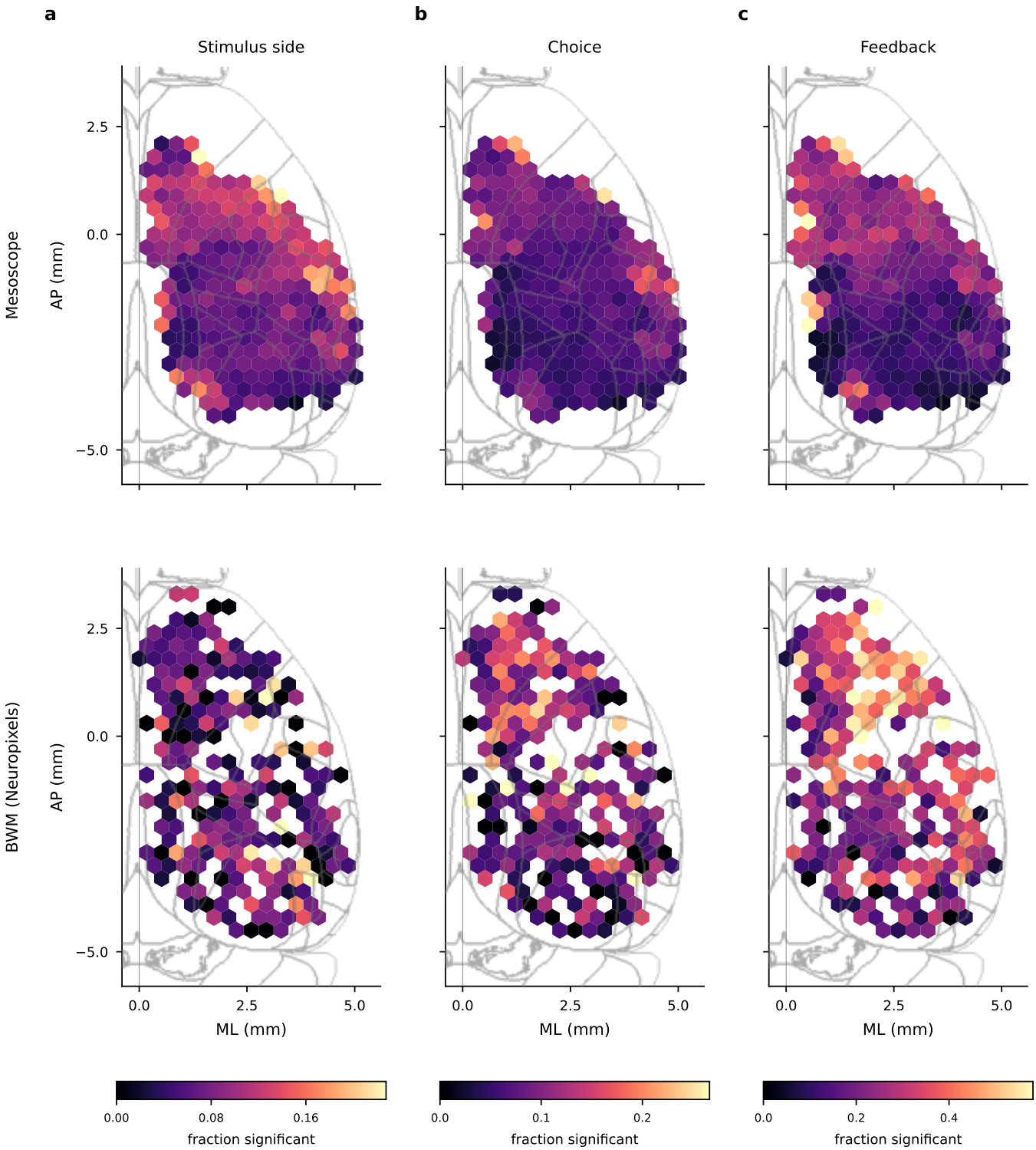



**Figure 1:** Dorsal cortex maps, mesoscope two-photon imaging (top row; deconvolved activity of isocortex neurons, canonical sessions) vs Brain Wide Map Neuropixels recordings (bottom row; good isocortex units, 2023 release). Each neuron is assigned to a hexagonal bin (0.4 mm vertex to vertex) by its medio-lateral (ML) and antero-posterior (AP) Allen CCF position relative to bregma; both hemispheres are folded onto the right one (ML $\rightarrow$ |ML|). Grey lines: Allen top-view region boundaries. Trials: choice made and reaction time 0.08–2 s (BWM criteria). Colour scales are shared between the two rows. Single-neuron tuning: the BWM single-cell test (brainwidemap single_cell_stats, re-implemented on the GPU and checked against the original), applied identically to both modalities to the mean activity in each window. Mann–Whitney $U$ statistics are computed within condition strata, summed and normalised; the null distribution comes from 3000 permutations of the condition labels within consecutive blocks, and $p$ is the rank of the observed statistic. Mesoscope windows are 0.3 s long because the frame period is about 0.2 s. **a** Fraction of neurons in each bin significantly tuned to stimulus side ($p < 0.05$; bins with $\geq$10 neurons pooled across sessions); left vs right stimulus, compared within block $\times$ choice; window $[0, 0.1]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to stimulus onset. **b** Same for choice (left vs right turn, within block $\times$ stimulus side); window $[-0.1, 0]$ s (BWM) or $[-0.3, 0]$ s (mesoscope) relative to movement onset. **c** Same for feedback (reward vs error, within block $\times$ stimulus side); window $[0, 0.2]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to feedback.